



Integrated modeling of urban water metabolism and ecosystem services: Indicators for the nexus of water, food, energy, and ecosystem services

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ABSTRACT

Today, cities are the main hubs of human activity, and this puts pressure on local natural resources and ecosystems, leading to challenges such as pollution and overuse. Employing comprehensive assessment and policymaking methods is effective in such areas due to the various components at play. For this purpose, the concept of the water–food–energy–ecosystem services (WFEES) nexus is used in this study. The components of this nexus are simulated through a newly developed tool that integrates urban water metabolism, water resources, and ecosystem services. This simulator evaluates management strategies and considers population growth and climate change scenarios to inform future policymaking. A novel composite index is developed to evaluate these strategies, considering economic, environmental, and social aspects. Additionally, this study utilizes the regional safe operating space (RSOS) framework to measure the state of the system compared with resilience thresholds. This study focuses on the western part of Tehran province and considers a 25-year simulation period (2016–2040). According to the findings, with the implementation of superior management packages, the average values of the composite index and the distance-to-threshold index will be 13.04 % and 38.03 % greater, respectively, than those in the business-as-usual (BAU) scenario. In conclusion, this study presents a set of indicators that urban policymakers can use to evaluate the state of the WFEES nexus under various climate and demographic scenarios. By examining how these indicators interact and evolve, the study provides valuable insights to guide effective and sustainable urban planning strategies for the future.

1. Introduction

Urbanization has significantly increased in the last century, and projections indicate that within the next 30 years, 65 % of the world's population will reside in cities (United Nations, 2018). Due to the increasing population growth in cities, these areas are struggling with problems such as resource scarcity and environmental pollution (Xu et al., 2023), which threaten sustainable urban development and the well-being of citizens (Shen et al., 2020).

Today, cities encompass diverse social, environmental, and infrastructural aspects. As complex social-ecological systems, urban areas

therefore pose a challenge for policymaking and planning. As a result, decision-makers need to develop comprehensive assessment frameworks to find trajectories that not only fulfill the needs and ensure the well-being of citizens but also lead to resource and environmental resilience (Asl-Rousta & Mousavi, 2025; Behboudian & Kerachian, 2021). Sustainable urban water management in arid and semi-arid regions depends on robust groundwater governance, particularly through establishing and enforcing water extraction thresholds. Conservation efforts, wastewater reuse, and leakage reduction alone are insufficient without strict groundwater extraction limits to prevent long-term depletion (Momeni et al., 2021; Moghaddasi et al., 2022). Moreover,

Abbreviations: WFEES, Water–food–energy–ecosystem services; RSOS, Regional safe operating space; BAU, Business-as-usual; CMUWMES, Coupled modeling of urban water metabolism and ecosystem services; SSP, Shared socio-economic pathway; GCM, Global climate model; LH-OAT, Latin-Hypercube One-factor-At-a-Time; R, Reliability; GEP, Gross ecosystem product; NPC, Nexus policy cost; UI, Urban intensity; WRE, Water reclamation efficiency; UME, Urban metabolism efficiency.

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effective national and regional policies to regulate water abstraction are critical, as inadequate enforcement has driven urban water supply systems toward a state of critical unsustainability (Banouei et al., 2015).

In recent years, the concept of the water–food–energy (WFE) nexus has been proposed as a framework for urban governance. Using this framework, water, food, and energy systems are evaluated as interconnected units, their interactions are assessed, and decision-makers are provided with a comprehensive picture of an urban area (Cristiano et al., 2021). Various methods, including input–output analysis (Deng et al., 2020; Tabatabaie & Murthy, 2021), life cycle assessment (Al-Ansari et al., 2015; Maarefi et al., 2024), system dynamics (Naderi et al., 2021; Ravar et al., 2020), and integrated indicators (Babaei et al., 2024; Mohammadpour et al., 2019), have been utilized to conduct analyses via the WFE nexus approach. Among these, urban metabolism—grounded in material flow analysis (MFA)—has been widely recognized for its capacity to quantify inputs, outputs, and internal interactions across urban subsystems, making it a practical tool for integrated resource and environmental management (Wang et al., 2020).

Several studies have demonstrated the integration of urban metabolism within the WFE nexus framework. Villarreal Walker et al. (2014) assessed the influence of various technologies on urban metabolism and the flows of nitrogen, phosphorus, carbon, water, and energy. Hu et al. (2019) combined urban metabolism with system dynamics to simulate the management of a multifunctional reservoir under uncertainty, addressing the competing demands of hydropower, agriculture, and urban water supply. Emamjomehzadeh et al. (2023) developed a model based on urban water metabolism within the WFE nexus framework to support integrated water resources management. Their model effectively assessed how different sectors interact and how various management scenarios impact water-related subsystems and their interconnections. The water–food–energy–environment nexus (Momblandh et al., 2019), the water–food–energy–climate–land nexus (Sušnik et al., 2021), and the water–food–energy–ecosystem nexus (Zhang et al., 2024) are all expanded frameworks that build on the original WFE (water–food–energy) nexus by incorporating additional interconnected concepts.

One of the concepts that has been considered in past studies regarding the WFE nexus has been ecosystem services (Sahle et al., 2019; van den Heuvel et al., 2020; Yuan & Lo, 2020). Ecosystem services are the benefits provided to human societies by the surrounding environment. These services play crucial roles in studies on natural resources and environmental management, as they are influenced by various natural factors and human activities (Behboudian et al., 2023; Khani et al., 2025). These services can also positively contribute to urban metabolism fluxes. Services like water yield, crop production, and bio-energy production directly enhance the supply of water, food, and energy. Urban cooling contributes to lower energy consumption, while carbon sequestration plays a vital role in storing carbon emissions resulting from water, food, and energy production processes (Ding et al., 2023; Elliot et al., 2022). Therefore, integrating these services with the WFE nexus in resources and environmental planning studies provides a more comprehensive view and allows for the consideration of interactions between these services and other sectors (Karabulut et al., 2019).

A clear limitation in existing studies is the absence of models that couple urban metabolism and ecosystem services in a comprehensive way. This leads to an incomplete understanding of how urban metabolism interacts with ecosystem services. Even recent frameworks calling for integration highlight this gap, emphasizing the need for studies that explicitly couple urban metabolism analysis with ecosystem services modeling (Cárdenas-Mamani & Perrotti, 2022, 2024).

Another topic that has received insufficient attention in nexus-based research is measuring resilience. While resilience is increasingly recognized as essential for sustainable resources management, it is rarely quantified in a way that enables decision-making (Quinlan et al., 2016). One way to measure resilience is by assessing the system's state against

regional and global thresholds (Dearing et al., 2014). The RSOS framework was developed for this purpose, focusing on regional thresholds (McLaughlin, 2018). However, current nexus-based assessments lack a mechanism to determine how close an urban system may be to breaching critical thresholds of water scarcity, energy security, or ecosystem degradation. This is a significant limitation for resilience analysis, as it prevents planners from identifying impending tipping points in urban systems.

This study presents a new approach to explore the following research question: How can coupled modeling of urban water metabolism and ecosystem services (CMUWMES) within the WFEES nexus framework support the development of effective management strategies to improve urban resilience and sustainability across different socio-environmental scenarios? Unlike previous studies that have analyzed resource flows and ecosystem benefits in isolation, this study simultaneously simulates urban water metabolism and ecosystem services dynamics. The proposed approach combines resources flows with ecosystem services such as green space provisioning and pollution purification, enabling analysis of relationships between the built environment and natural systems. In addition, the framework evaluates multiple future drivers simultaneously, including climate change, population growth, and land use change scenarios, providing a more comprehensive scenario analysis than previous studies that have examined limited factors. Another key contribution of this research is the introduction of novel indicators that balance environmental, economic, and social factors in evaluating WFEES performance. This study develops a composite index that integrates these three dimensions, offering a more holistic metric for assessing sustainability and resilience. For the first time, the RSOS framework is also employed as another criterion to measure the status of a WFEES nexus concerning regional thresholds. The insights gained from this research have direct implications for urban planning and management. By evaluating water, food, energy, and ecosystem services collectively, the model identifies integrated solutions that provide co-benefits across sectors rather than isolated sector-based improvements. Previous studies recommend improvements in a single sector without considering the impact on other sectors, whereas this coupled analysis highlights management strategies that optimize multiple outcomes simultaneously. Specifically, the findings of this study identify infrastructure projects and actions that should be prioritized for land management, waste and pollutants control, and water and energy consumption management.

The rest of the paper is structured as follows: Section 2 describes the study area, collected data, and proposed methodology. The results obtained from the proposed methods are presented in Section 3, followed by a discussion in Section 4. Section 5 provides concluding remarks and suggestions for future studies.

2. Materials and methods

2.1. Description of the study area

The Kan River basin is located west of Tehran city, the capital of Iran (Fig. 1). This area includes Tehran's urban districts 5, 9, 18, 21, and 22, along with the cities of Eslamshahr, Chahardangeh, and Kahrizak. The region is situated south of the Alborz Mountains and is characterized by an average temperature of 17 °C and an annual precipitation rate of 300 mm. This area includes water bodies, urban areas, agricultural lands, urban green spaces, and barren landscapes. According to the last official census in 2016, approximately 1720,000 people reside in this region (Sadri-Shojaei et al., 2025).

The region has various water resources. Water is provided to multiple sectors via the Karaj and Taleghan Dams, surface streams such as the Kan River, as well as groundwater and treated wastewater. The area also has various facilities, such as water treatment plants, wastewater treatment plants, aquifer extraction wells, and sewage wells, which are used for water and wastewater circulation. Water distribution networks

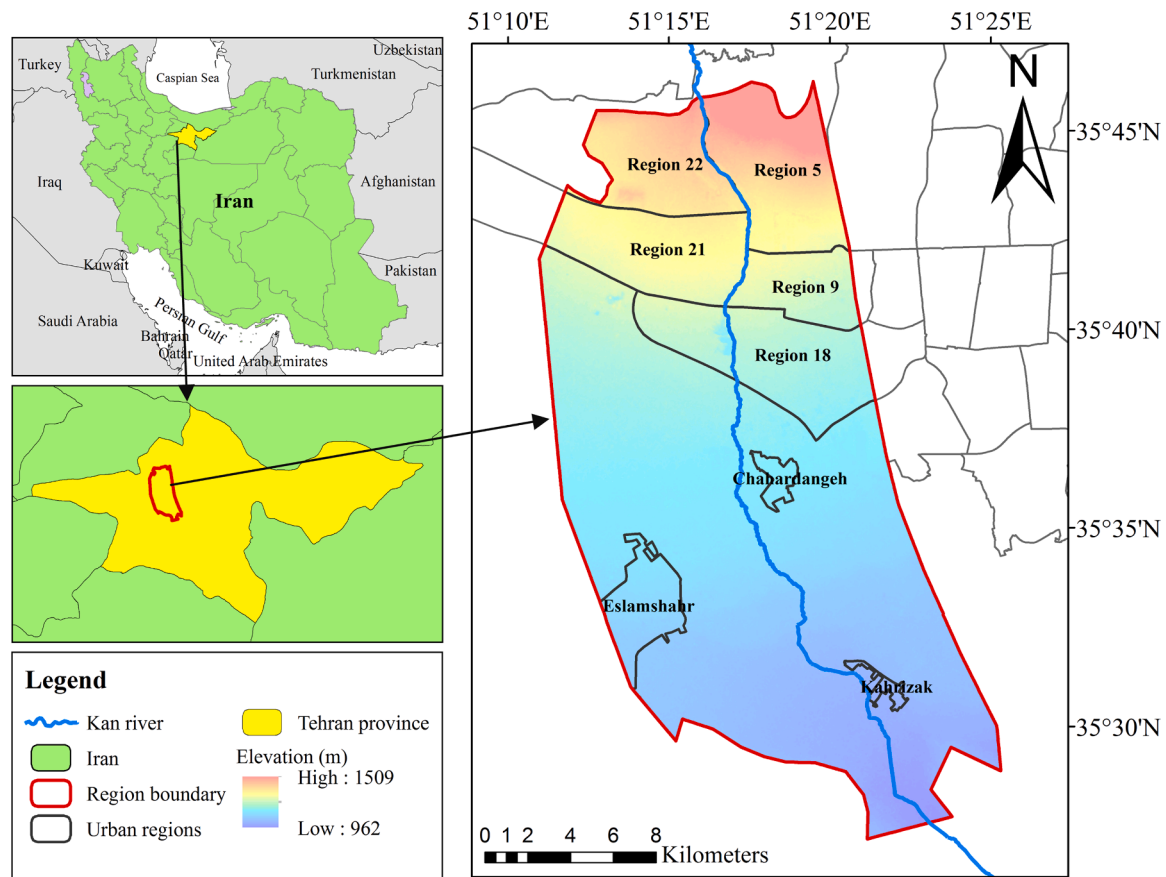


Fig. 1. The location of the study area in the western part of Tehran province in Iran.

in this region currently have a leakage rate of approximately 24 %. According to the leakage control plan, it is expected that by 2037, leakage will decrease to 10 % and stabilize at that level (Emamjomehzadeh et al., 2023).

Various end users, such as citizens, farmers, companies, and municipalities, depend on water resources in this region and, in some cases, compete for a greater share. Additionally, several issues have significantly impacted water resources and the ecosystem of this region. These include the quantitative decline in groundwater resources and subsidence, the discharge of untreated sewage into water resources, and the decline in water quality. Other contributing factors are population growth, increasing demand for resources, the reduction in the area of green spaces, high energy consumption, and rising greenhouse gas emissions (Sadri-Shojaei et al., 2025).

2.2. System identification and data collection

Fig. 2 presents the flowchart outlining the methodology steps proposed in this study. The initial step involves identifying the components of the system to be evaluated within the context of the WFEES nexus framework. This nexus consists of four main parts in this study. The water resources of the study area include surface flows, groundwater, and urban wastewater, which can be recycled in the system if treated. The supply, treatment, and distribution of these resources are carried out in the urban water metabolism sector. These processes consume a certain amount of materials and energy, and generate greenhouse gas emissions and waste. Ecosystem services constitute another component of this nexus. In addition to supporting water resources, these services can benefit the urban water metabolism sector by sequestering greenhouse gases and producing bioenergy. These services also have economic and welfare value for society. The social sector, which

encompasses end users such as citizens, farmers, and companies, relies on the three aforementioned sectors. Furthermore, the policies determined by this sector influence future management practices. Fig. 3 shows the various components of the WFEES nexus examined in this study using a conceptual model.

The next step is to gather the data required for the simulation. Various types of data were collected from different sources to simulate each of the aforementioned parts of the integrated system and to set up the models. Table S1 in the online supplementary material lists these data along with their source and scale.

2.3. Scenarios and strategy packages

In this study, the future climate situation is simulated under three climate scenarios from the set of shared socio-economic pathways (SSPs). Using global climate models (GCMs) outputs, Sadri-Shojaei et al. (2025) downscaled three climate scenarios for this region, SSP1–2.6, SSP2–4.5, and SSP5–8.5, which are used in this study. They also developed four land use change scenarios for changes to be achieved by 2040. Table S2 in the online supplementary material provides the land use figures for both the initial year and the final year of the simulation period for each of the four scenarios. Since the modeling in this study is on an annual scale, annual land use changes between the first year and the last year of the simulation period are calculated via the linear interpolation method, which previous studies have employed (Aghsaie et al., 2020; Wagner et al., 2016). Additionally, scenarios related to population growth in the study area are examined. This study formulates predictive functions for estimating future population growth, utilizing historical trends derived from official census data within the study area for the years 2006, 2011, and 2016. Table S3 in the online supplementary material shows these functions developed for each region of the

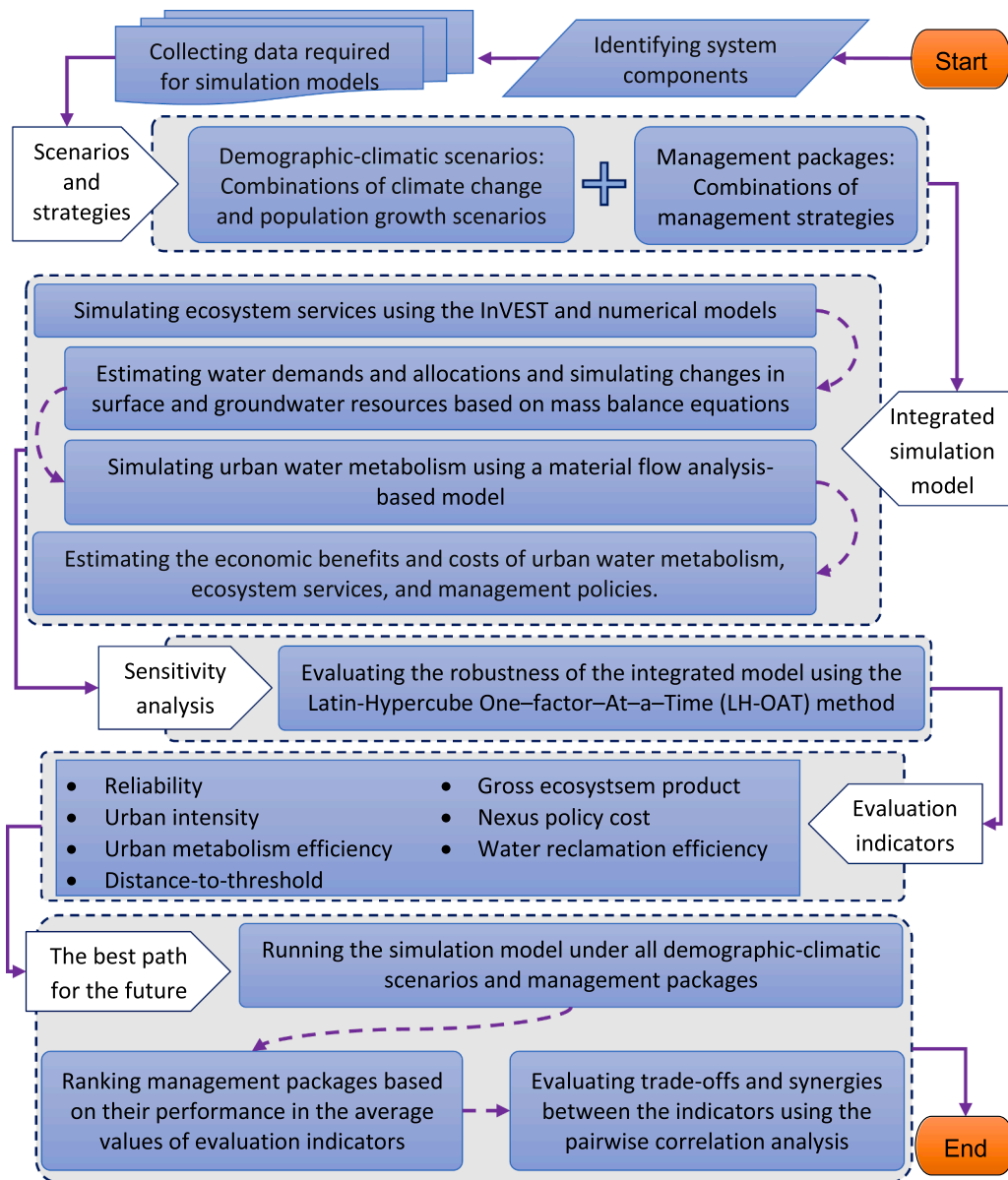


Fig. 2. A flowchart of a proposed methodology for planning for the water–food–energy–ecosystem services nexus under the impacts of natural and human factors.

study area. These functions are designed to project future demographic changes under the assumption that current trends will persist, representing the BAU population scenario. In addition to this scenario, other scenarios have also been developed, respectively considering 10 % more and 10 % less than these projections each year.

Furthermore, this study presents strategy packages based on different combinations of management measures and infrastructure projects. The impacts of these packages on different system components are evaluated during the simulation period. Strategies include urban water consumption management, cropping pattern management, green space vegetation management, wastewater collection and treatment network development, land use management, rainwater harvesting, and bioenergy production.

2.4. Water–food–energy–ecosystem services nexus

In this study, water resource simulation and allocation models, a water demand estimation model, a water metabolism model based on material flow analysis, and ecosystem services simulation models are integrated to simulate the components of the WFEES nexus. Fig. A1 in

the appendix shows the sub-sections of the integrated simulator and their inputs and outputs. The following sections elaborate on each of these components.

2.4.1. Ecosystem services simulation

According to MEA (2005), ecosystem services are generally divided into four categories: 1) provisioning, 2) regulating, 3) supporting, and 4) cultural. This paper simulates and evaluates seven ecosystem services (Table 1). Some of these services (annual water yield, urban cooling, stormwater retention, and carbon sequestration) are simulated with models rewritten from the InVEST software (Sharp et al., 2020), whereas others (recreational spaces, agricultural production, and bioenergy production potential) are simulated with models developed by Sadri-Shojaei et al. (2025). Table S4 in the online supplementary material presents the equations used in these models along with their details.

2.4.2. Water resources simulation and allocation

This study simulates water resources using models by Darbandsari et al. (2020), which are based on the mass balance and Darcy's law and

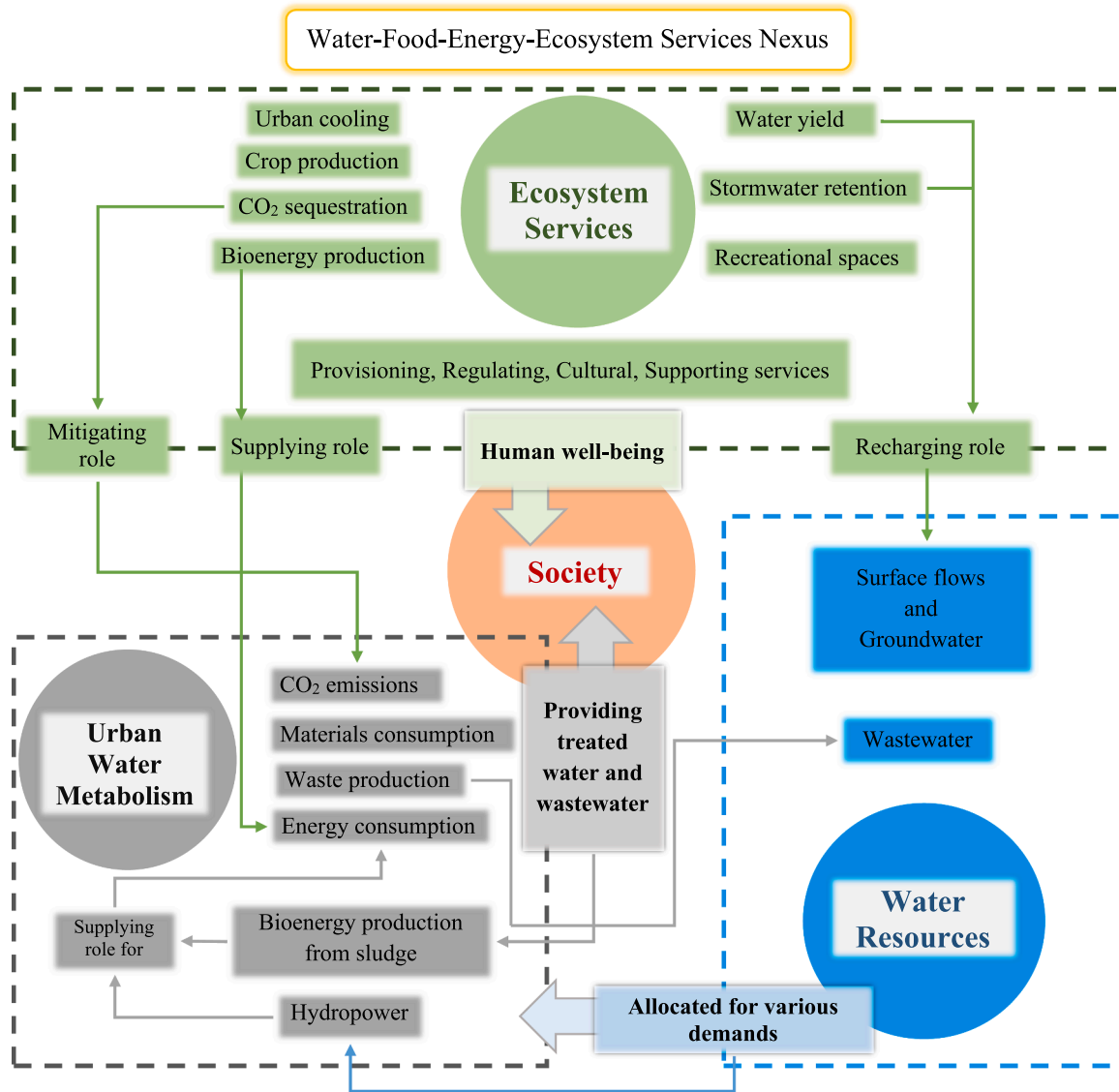


Fig. 3. A conceptual presentation of the components of the water-food-energy-ecosystem services nexus and their relationships.

Table 1

A summary of ecosystem services assessed in this study.

Ecosystem service	Description
Potential bioenergy production	The amount of energy that can be obtained from agricultural residues.
Carbon sequestration	The amount of CO ₂ stored in carbon pools in different land uses.
Crop production	The volume of crops produced on agricultural lands.
Recreational spaces	The area of green spaces accessible to people living in the area.
Stormwater retention	The volume of precipitation that is converted into evapotranspiration or percolates into the aquifer.
Urban cooling	The effect that vegetation cover has on reducing air temperatures in urban areas.
Water yield	The volume of surface flows that occur due to precipitation.

consider surface and groundwater interactions. The calculations related to the conversion of precipitation into runoff or other components of the water balance equation are explained in the ecosystem services section. Groundwater quality is modeled using the approach by Khorasani et al. (2020), which uses nitrate concentration as an indicator. This model applies the mass balance equation and accounts for inflows and

withdrawals. Additionally, a water resource allocation model has been used that operates based on the prioritization of different end users. Emamjomehzadeh et al. (2023) developed this model for allocating water from different resources to end users. Table S5 in the online supplementary material gives information about this prioritization.

2.4.3. Demand estimation

The distribution of water resources to various sectors is determined by the prioritization outlined in the previous section and the projected water demands each year. The urban sector water demand is calculated based on daily water consumption per capita. Farmers in this region are significant end users of water, with their demand estimated on the basis of the needs of the primary crops, such as wheat, barley, and alfalfa (Sadri-Shojaei et al., 2025). Due to insufficient data, the water demand for artificial Chitgar Lake, gardens, and industry is assumed to have fixed values of 2.18, 5, and 20 million cubic meters (MCM) per year, respectively. Costello (1994) developed the Water Use Classifications of Landscape Species (WUCOLS) method to estimate the water demand of green spaces, which is also used in this study.

2.4.4. Urban water metabolism simulation

Emamjomehzadeh et al. (2023) developed an urban water

metabolism model that simulates energy, material, and greenhouse gas fluxes by considering the circulation of water and wastewater between resources, facilities, and end users. In the energy flux calculation section, this model assesses several elements: the energy consumed for transferring water, the energy used for treating water and wastewater, the energy required to maintain urban green spaces, the energy consumed by machinery in the agricultural sector, and the energy generated through biogas production in wastewater treatment plants. In the greenhouse gas emissions estimation section, in addition to calculating emissions due to energy consumption in the mentioned cases, emissions from sewage wells and the activated sludge process in wastewater treatment plants are also considered. The other component of this model focuses on the material flux within the system. This section calculates the quantities of materials required for water and wastewater treatment, the amount of fertilizer and agricultural seeds needed, the fertilizer produced from sewage sludge, and the pesticides required by farmers. In addition to the amount of materials, embodied energy, and greenhouse gas emissions are also calculated. Table S6 in the online supplementary material presents the equations used in different parts of this model.

2.4.5. Sensitivity analysis

To evaluate the robustness of the integrated model developed in this study, a sensitivity analysis is conducted on its inputs. The Latin-Hypercube One-factor-At-a-Time (LH-OAT) method, proposed by van Griensven et al. (2006), is suitable for sensitivity analysis in models with multiple inputs and outputs. This method determines the sensitivity of variables and parameters using Eq. (1):

$$S_{ij} = \left| \frac{\left(\frac{M(e_1, \dots, e_i \times (1+f_i), \dots, e_p) - M(e_1, \dots, e_i, \dots, e_p)}{[M(e_1, \dots, e_i \times (1+f_i), \dots, e_p)] + M(e_1, \dots, e_i, \dots, e_p)} \right)}{f_i} \right| \times 100 \quad (1)$$

where S_{ij} represents the relative change in the j th output of model M due to a change in the i th input (e_i). The term f_i refers to the rate of change in the i th input, while p indicates the total number of inputs in the model. After the average relative change for each input (S_i) is calculated, the model's sensitivity to each variable and parameter can be assessed.

2.5. Evaluation indicators

A variety of indices are employed to evaluate the system from multiple perspectives and to inform policy decisions for the future. The values of these indices are derived from the outputs of simulation models. Table 2 presents a concise overview of these indices, followed

Table 2
Proposed indices for evaluating the state of the WFEES nexus.

Index	Related Sector(s)	Purpose
Reliability	Water, Energy, Ecosystem Services	Measuring water, energy, and carbon sequestration supply against existing demand
Gross ecosystem product	Ecosystem Services, Economy	Assessing the status of ecosystem services and their financial benefits
Nexus policy cost	Economy, Water, Energy, Food	Estimating the financial cost of implementing each management action and infrastructure project
Urban intensity	Land, Ecosystem Services	Assessing land use changes against urbanization growth
Water reclamation efficiency	Water, Energy	Evaluating the efficiency of the water recycling system
Urban metabolism efficiency	Water, Energy, Food	Assessing the efficiency of urban resource flows to increase production while minimizing waste

by a detailed explanation of the calculation methods.

2.5.1. Normalization

To evaluate and compare various indicators effectively, it is essential to adopt a normalization process. This study utilizes the Max-Min normalization method, as delineated in Eqs. (2) and (3).

$$x' = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (2)$$

$$x' = \frac{x_{\max} - x_i}{x_{\max} - x_{\min}} \quad (3)$$

In the above equations, x_i represents the i th value of the data series related to an indicator, and x' is the normalized value. Also, $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the data series related to that indicator, respectively. Eq. (2) applies to indicators with a positive effect, whereas Eq. (3) applies to those with a negative effect (Qin et al., 2022).

2.5.2. Reliability index

The reliability (R) index is based on the supply-demand ratio per year (Eq. (4)). This study calculates the index by evaluating three key ratios: the ratio of available water to water demand, the ratio of energy production to energy demand, and the ratio of carbon sequestration to carbon emissions.

$$R = \frac{1}{n} \sum_{i=1}^n \frac{S_i}{D_i} \quad (4)$$

In the above equation, S_i denotes the supply and D_i denotes the demand for the i th ratio, while n represents the total number of ratios.

2.5.3. Gross ecosystem product index

This index allows us to measure the contribution of the environment to economic development. The concept of gross ecosystem product (GEP) is derived from the valuation of ecosystem services and goods (Hao et al., 2022). The total monetary value of ecosystem provisioning services, ecosystem regulating services, and ecosystem cultural services is measured annually in a given area using this index.

$$GEP = \sum_{i=1}^n EP_i \times P_i + \sum_{j=1}^m ER_j \times P_j + \sum_{k=1}^l EC_k \times P_k \quad (5)$$

In Eq. (5), EP_i , ER_j , and EC_k represent the measured values of the i th provisioning service, j th regulating service, and k th cultural service, respectively. Additionally, n , m , and l represent the total numbers of provisioning, regulating, and cultural ecosystem services, respectively. Furthermore, P_i , P_j , and P_k are the unit monetary values of the i th provisioning service, j th regulating service, and k th cultural service, respectively (Linbo et al., 2022). In this study, different methods are utilized to value services, and Eq. (6) is also used to consider the effect of the inflation rate on the value of these services during the simulation period.

$$P_i = P_1 \times (1 + inf)^i \quad (6)$$

In the equation above, P_i denotes the price of a good or service in the i th year, while P_1 indicates the price of the same good or service in the base year. Additionally, inf represents the annual inflation rate. Table 3 presents the types of ecosystem services, their valuation method, their unit monetary value in the base year, and the source of the monetary value.

2.5.4. Nexus policy cost index

The implementation of each management strategy has an economic cost in addition to its impact on natural resources and the environment. The nexus policy cost (NPC) index calculates the total annual cost of implementing the strategies listed in Table S4 in the online

Table 3
Details of the valuation of ecosystem services in this study.

Service Category	Service	Valuation method	Unit monetary value in the base year (2016)	Source
Ecosystem provisioning services (EPV)	Water yield	Based on the volume of surface runoff and the price of water consumed in the agricultural sector	0.011 (USD/m ³)	(Skardi et al., 2020)
	Crop production	Based on the estimated price of each crop	Wheat: 0.314 (USD/kg), Barley: 0.248 (USD/kg), Alfalfa: 0.19 (USD/kg)	(Darbandsari et al., 2020)
Ecosystem regulating services (ERV)	Potential bioenergy production	Based on electricity price	0.0742 (USD/kWh)	(Skardi et al., 2020)
	Carbon sequestration	Based on the global carbon price	13.96 (USD/ton)	(Shojaie Shami et al., 2021)
	Stormwater retention	Based on the volume of water infiltration into the aquifer and the price of groundwater for the urban sector	0.0865 (USD/m ³)	(Skardi et al., 2020)
	Urban cooling	Based on the amount of energy consumption reduction due to urban cooling, which is calculated using the method introduced by Sedaghat and Sharif (2022), and the price of electricity	0.0742 (USD/kWh)	(Skardi et al., 2020)
Ecosystem cultural services (ECV)	Recreational spaces	Based on the area of urban green spaces and the travel cost method	41.21 (USD/ha)	(Amirnejad & Ataie Solout, 2021)

supplementary material.

$$NPC = \sum_{i=1}^n Cost_i \quad (7)$$

In Eq. (7), $Cost_i$ is the annual value of the i th cost indicator, and n is the total number of cost indicators.

One of the important infrastructure projects in the study area is the construction of urban wastewater treatment plants. The overall cost for these projects encompasses both the initial investment and the ongoing operating expenses. Abbasi et al. (2021) discussed these costs in a study of a wastewater treatment plant located in Tehran. Their estimate indicates that the annual cost for constructing and operating a wastewater treatment plant using the activated sludge process is 1.286 USD/m³. In this study, it is assumed that all wastewater treatment plants, whether built or under construction, operate using the activated sludge process. Additionally, bioenergy production from biogas generated in wastewater treatment plants is a viable infrastructure project with significant implementation potential. In addition to contributing positively to the energy supply, this project also helps reduce methane emissions from sludge generated in treatment plants. Only the southern Tehran wastewater treatment plant, located outside the study area, can produce energy from biogas. According to information provided by the Tehran Province Water and Wastewater Company, the total annualized project cost of this part of the treatment plant, which processes 450 tons of sludge daily (Tabesh et al., 2019), is 3.068 USD/ton.

Another management strategy evaluated in this study is the implementation of rainwater harvesting systems. These systems have the potential to supply water for unconventional applications, thereby alleviating the community's demand for other water resources. The method for estimating the size of the required tanks for these systems is detailed in the online supplementary material. Saeedi and Goodarzi (2020) presented the costs of setting up and operating these systems in a study. They estimated that the annualized cost of purchasing and installing tanks, based on their capacity, would be 19.4 USD/m³, with an annual operating cost of 0.09 USD/m³. Water management in the domestic sector affects the cost of water treatment, and reducing demand in this sector results in less material and energy being used in water treatment plants in the region. The water treatment cost per cubic meter for the domestic sector is worked out by calculating the embodied energy of the materials used and the electrical energy required to operate the water treatment facilities. The demand for water from other sectors besides the urban sector also influences the volume of groundwater extracted, thereby affecting the energy consumption of the pumps that are needed. Since the costs of pumping water in order to meet the needs of each sector differ, Table A1 in the appendix shows the cost of pumping

each cubic meter of water per meter for each sector. Using these figures, the volume of water needed by each sector, along with the distance from the groundwater level to the end user's location, can be used to calculate the costs of water transfer.

Other costs include the cost of maintaining green spaces and the cost of supplying agricultural fertilizers. According to the statistics of the Ministry of Agriculture-Jahad of Iran in 2016 (base year), the average price of chemical fertilizers was 25.15 USD/ton. Based on Eq. (4), the annual price is estimated on the basis of the price in the base year and the inflation rate, and depending on the amount of fertilizer consumed, the total cost of the agricultural sector to supply fertilizers is calculated. The cost of maintaining green spaces is calculated on the basis of energy consumption, which the metabolism model determines, and the annual price of electricity.

2.5.5. Urban intensity index

This index evaluates the level of urbanization and residential land expansion in an area and helps policymakers with urban planning. Population growth and land use change are the most important factors affecting this index (McMahon & Cuffney, 2000).

$$UI = \frac{1}{n} \sum_{i=1}^n u'_i \quad (8)$$

In the above equation, u'_i is the normalized value of the i th indicator, and n is the total number of indicators used to calculate the UI index. Indicators such as population density (person/m²), land development (%), and the proportion of arable land (m²) are used in this study. The population density and land development indicators contribute positively to the urban intensity index, thereby increasing its overall value. In contrast, a greater proportion of arable land negatively impacts the urban intensity index.

2.5.6. Water reclamation efficiency index

Since wastewater treatment facilities play an important role in water resources planning in urban areas, their efficiency can be considered an index for comparing management policies. This study measures the water reclamation efficiency (WRE) index using two indicators: the percentage of wastewater treated per year and the net energy consumed in the facility for treating each cubic meter of wastewater.

$$WRE = \frac{1}{n} \sum_{i=1}^n wr'_i \quad (9)$$

In the above equation, wr'_i is the normalized value of the i th indicator, and n represents the number of indicators under consideration. Notably, the percentage of wastewater treated contributes positively to the calculation of this index, whereas the net energy consumed has a

negative impact.

2.5.7. Urban metabolism efficiency index

The urban metabolism efficiency (UME) index is used to evaluate and track system flows. A higher value of this index means better resource use and reduced waste. Six indicators derived from the outputs of the newly developed model are used to calculate this index (Table 4).

$$UME = \frac{1}{n} \sum_{i=1}^n um'_i \quad (10)$$

In the above equation, um'_i represents the normalized value of the i th indicator. Also, n denotes the total number of indicators that have been assessed to calculate the UME index.

2.5.8. Composite index

This study uses the indices defined in the previous sections to develop a novel composite index. This composite index evaluates the state of the WFEES nexus by considering economic, social, and environmental aspects. To develop this index, it is essential to first normalize the values of all indices using Eqs. (2) or (3). Among the proposed indices, the nexus policy cost and urban intensity indices exert a negative influence, whereas the remaining indices contribute positively to the values of the composite index.

Subsequently, the indices are weighted and integrated by employing the linear weighted aggregation method. This study uses the entropy method, which is an objective weighting method, to determine the weights of indices. In this method, a lower entropy value for an index indicates that it carries more information and has greater weight compared to other indices (Li et al., 2024). This method calculates the weights of the indices according to Eqs. (11) and (12):

$$H_i = -\frac{1}{\ln m} \sum_{j=1}^m \frac{r_{ij}}{\sum_{j=1}^m r_{ij}} \ln \frac{r_{ij}}{\sum_{j=1}^m r_{ij}} \quad (11)$$

$$w_i = \frac{1 - H_i}{n - \sum_{i=1}^n H_i} \quad (12)$$

where r_{ij} is the j th normalized value of the data series for the i th index, and n and m represent the total number of indices and their measured values, respectively. Also, w_i and H_i represent the weight and the information entropy of the i th index, respectively. After the weights are determined, the composite index (CI_p) is computed via Eq. (13):

$$CI_p = \frac{1}{n} \times \frac{1}{m} \sum_{s=1}^n \sum_{y=1}^m \sum_{i=1}^k \left(w_i \times p_{iys}^p \right) \quad p = 1, 2, \dots, z \quad (13)$$

where n represents the number of demographic-climatic scenarios, m is the number of simulation time steps, k is the number of indices, and z is the number of management packages. Also, w_i represents the weight of the i th index, and p_{iys}^p represents the normalized value of the i th

indicator at the y th time step and under the demographic-climatic scenario s and management package p .

2.5.9. Regional safe operating space

In addition to the composite index values, this study employs the RSOS framework to assess the effects of management packages on system components. This framework measures system performance against boundaries derived from regional and global standards and thresholds (McLaughlin, 2018; Zhang et al., 2023). Sadri-Shojaei et al. (2025) defined indicators and their associated thresholds for assessment within this framework for the same study area as this study. Table A2 in the appendix presents these indicators and defined thresholds.

To assess the system's state, the values of the indicators (x_{ij}) are measured at each time step of the simulation (j). These values are compared against the established thresholds (L_j), utilizing Eqs. (14) (for the minimum threshold) and (15) (for the maximum threshold).

$$d_{ij} = x_{ij} - L_j \quad (14)$$

$$d_{ij} = L_j - x_{ij} \quad (15)$$

Next, the values calculated via the above equations are normalized and linearly integrated according to Eq. (16) to evaluate the overall state of the system at each time step using this framework:

$$D_i = \frac{1}{n} \sum_{j=1}^n d'_{ij} \quad (16)$$

where n denotes the total number of indicators and D_i describes the overall state of the system compared to the thresholds at the i th time step. Also, d'_{ij} represents the normalized value of the j th indicator at the i th time step. The more the value of the definition index in the above equation tends toward negative values, the more thresholds the system crosses.

At the end of the simulations and calculation of the indices values for each management package, these packages are compared with each other based on the values of the composite index and the distance index. The package that demonstrates better performance than the alternatives in both cases will be selected as the superior management package. If it is not possible to identify a single best package, the Pareto front of non-dominated options will be presented.

2.6. Trade-off and synergy analysis

An analysis of trade-offs and synergies is essential for understanding the relationships between the indicators defined in the previous section and how management policies influence these interactions. This study employs the pairwise correlation analysis method, which is commonly used in similar studies (Kashef et al., 2025; Li et al., 2024; Wu et al., 2022), to assess the relationships between indicators. The correlation between the indicators is calculated using the Pearson correlation coefficient (Eq. (17)).

$$r(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (17)$$

In the above equation, $r(X, Y)$ is the Pearson correlation coefficient between variables X and Y . x_i and y_i are the i th values of the two variables, X and Y , respectively. n represents the total number of data points for each variable, while $\bar{x}$ and $\bar{y}$ denote the average values for X and Y , respectively. The values of this coefficient indicate the type of relationship between the indicators. Negative values denote a trade-off between two indicators, whereas positive values indicate a synergistic relationship. If the value of the correlation coefficient is zero, it signifies a neutral relationship between the two indicators (Jopke et al., 2015).

Table 4

Indicators related to the evaluation of the urban metabolism efficiency index.

Indicator description	Unit	Impact
The ratio of treated wastewater reuse to total water consumption	%	(+)
The ratio of fertilizer from sludge produced in wastewater treatment plants to total fertilizer consumed in the agricultural sector	%	(+)
The ratio of carbon emissions from water supply and wastewater treatment facilities to total water consumption	ton/m ³	(-)
The ratio of bioenergy production in wastewater treatment plants to total energy consumption in wastewater treatment plants	%	(+)
The amount of untreated wastewater per population living in the study area	m ³ /person	(-)
The amount of agricultural production per population living in the study area	ton/person	(+)

3. Results

3.1. Climate, population, and management scenarios

Three climate change scenarios (i.e., SSP1–2.6, SSP2–4.5, and SSP5–8.5) have been employed to project annual changes in climate variables, including precipitation and minimum and maximum air temperatures. By integrating these scenarios with three population growth projections, nine potential futures for the system evaluated in this study have been established. Information about these scenarios is presented in Table 5.

In addition to the demographic–climatic scenarios, 13 packages of different management strategies and solutions have been developed, and their impacts on the various components of the WFEES nexus and evaluation indicators have been evaluated. This comprehensive analysis enables us to evaluate the effects of different combinations of management actions, facilitating informed planning for the future and guiding us toward the identification of optimal policy pathways. Table 6 provides details of these packages.

3.2. Model performance and sensitivity analysis

Figure S1 in the online supplementary material compares the outputs of the newly developed model for the validation period from 2011 to 2016 with the observed data. The results indicate that the root mean square error (RMSE) value for the model outputs related to surface outflow was 5.6 MCM. The RMSE values for the outputs related to changes in the water table levels for zone 1, zone 2, and zone 3 of the aquifer were 0.9, 0.9, and 0.7 m, respectively. Also, the RMSE values for the other model outputs, such as surface runoff and actual evapotranspiration, were 1.7 and 5 MCM, respectively, compared with the observed data.

The LH–OAT method was employed to identify the sensitive inputs. Table S7 in the supplementary materials online presents the results of the sensitivity analysis for various inputs. Accordingly, 10 inputs, including initial land use, wastewater treatment percentage, land use change scenario, economic inflation rate, volume of transferred water from dams, aquifer storage coefficient, population, surface flow volume in the West Flood diversion channel, and climatic variables (precipitation and air temperature), are the most sensitive variables and parameters of the integrated simulator.

3.3. Simulation of the strategies and scenarios

The integrated simulator model was run from 2016 to 2040 using annual time steps to evaluate the system's status under each strategy package and demographic–climatic scenario. Notably, between 2016 and 2020, as a baseline period, the simulator was run only under the BAU scenario, and the effects of demographic–climatic scenarios and management strategies were not considered.

The evaluated packages were ranked based on their average performance during the simulation period and across all demographic–climatic scenarios (details in Table S8 in the online supplementary material). The performance of these packages was measured based on

the average of the composite and distance-to-threshold indices. Fig. 4 shows the state of each package concerning these two indices; in particular, packages 6 and 9 performed better than the other options and were non-dominated. On the other hand, packages P2 and P11 had the worst performance among the options considered. Following these two packages, the BAU management scenario, represented by package P1 along with packages P13 and P4, also showed poor performance.

Fig. 5 illustrates the weights of the sub-indices used to calculate the composite index and provides details on the performance of the leading packages compared to the BAU scenario across the defined sub-indices and indicators. This figure indicates that the WRE sub-index received the highest weight (0.22), whereas the GEP sub-index received the lowest weight (0.05) according to the entropy method.

The calculated values for the sub-indices associated with the composite index, when packages P9 and P6 were applied, were nearly equal. In contrast, the values for package P1 were generally lower, except for the NPC sub-index, which pertains to the economic cost of management measures. Specifically, due to the non-implementation of these projects in package P1, the value of this index has been higher, which indicates a lower economic cost (due to the negative effect of this index and using Eq. (2) to normalize its values). Regarding the regional safe operating space indicators, the performance of the P9 and P6 packages was similar, with the only significant difference being the green space per capita indicator (GS), where the average value was higher for P9. Most of the average values recorded for the P1 package were either lower than or equal to those of the higher-tier packages. However, P1 showed higher values for three indicators: groundwater level in aquifer zone 1 (GW1), available freshwater (AFW), and built-up area (BA).

Fig. 6 shows the annual changes in the values of the distance and composite indices during the simulation period under the implementation of the management packages and demographic–climatic scenarios. According to the results, during the base period, the values of both indices decreased. Based on Fig. 6a and b, if the current trend persists (package P1), the distance index will experience an increasing trend, and the value of the composite index will decline after experiencing an increasing trend in the first 10 years of the policymaking period and will remain almost constant until the end of the simulation period. The increasing trend of the values for the two indices is more pronounced under the superior management packages P9 and P6 than under the P1 package. The P9 package outperforms P6 in terms of the distance index, whereas P6 outperforms P9 in terms of the composite index. The distance index values for package P1 were 44.97 % and 31.09 % lower, on average, than those of packages P9 and P6, respectively.

Additionally, the average calculated values for the composite index under P1 are 12.06 % and 14.02 % lower than those under P9 and P6, respectively. According to Fig. 6c and d, which illustrate the annual changes in the two indices under the BAU management scenario and the demographic–climatic scenarios, the distance index is sensitive to both the climatic and demographic components of these scenarios. In contrast, the composite index has demonstrated sensitivity only to demographic aspects. In the initial years, the effects of these scenarios for both indices were more distinct, while in the later years, the calculated values became increasingly similar.

Table 5

Projections under demographic–climatic scenarios in the study area.

Population growth		Climate change			
Scenarios (Code)	Population in 2040	Scenarios (Code)	Average annual precipitation (mm)	Average maximum air temperature (°C)	Average minimum air temperature (°C)
10 % less than the BAU (D1)	1942,693	SSP1–2.6 (CC1)	205	24.4	14.4
BAU (D2)	2158,548	SSP2–4.5 (CC2)	227	24.5	14.5
10 % more than the BAU (D3)	2374,402	SSP5–8.5 (CC3)	226	24.2	14.3

Table 6
Details of the management packages examined in this study.

Package name	Management strategies									
	Land use change	Water consumption management in the agricultural sector	Water consumption management in urban green spaces	Irrigation efficiency	Progress of the wastewater collection network project	Water consumption management in the urban sector	Type of wastewater treatment system	Artificial groundwater recharge	Rainwater harvesting	Bioenergy production in wastewater treatment plants
P1 (Business as usual)	Business-as-usual	No change	No change	Not improved	Slow progress	No reduction	Centralized	Not implemented	Not implemented	Not implemented
P2	Business-as-usual	Cropping pattern with low and medium water demand	Vegetation cover with low and medium water demand	Improved	Slow progress	Consumption control	Centralized	Not implemented	Not implemented	Not implemented
P3	Business-as-usual	No change	No change	Not improved	Fast progress	No reduction	Decentralized	Implemented	Implemented	Implemented
P4	Business-as-usual	Cropping pattern with low and medium water demand	Vegetation cover with low and medium water demand	Improved	Fast progress	Consumption control	Decentralized	Implemented	Implemented	Implemented
P5	Urban expansion	Cropping pattern with low and medium water demand	Vegetation cover with low and medium water demand	Improved	Slow progress	Consumption control	Centralized	Not implemented	Not implemented	Not implemented
P6	Urban expansion	No change	No change	Not improved	Fast progress	No reduction	Decentralized	Implemented	Implemented	Implemented
P7	Urban expansion	Cropping pattern with low and medium water demand	Vegetation cover with low and medium water demand	Improved	Fast progress	Consumption control	Decentralized	Implemented	Implemented	Implemented
P8	Balanced development	Cropping pattern with low and medium water demand	Vegetation cover with low and medium water demand	Improved	Slow progress	Consumption control	Centralized	Not implemented	Not implemented	Not implemented
P9	Balanced development	No change	No change	Not improved	Fast progress	No reduction	Decentralized	Implemented	Implemented	Implemented
P10	Balanced development	Cropping pattern with low and medium water demand	Vegetation cover with low and medium water demand	Improved	Fast progress	Consumption control	Decentralized	Implemented	Implemented	Implemented
P11	Green space development	Cropping pattern with low and medium water demand	Vegetation cover with low and medium water demand	Improved	Slow progress	Consumption control	Centralized	Not implemented	Not implemented	Not implemented
P12	Green space development	No change	No change	Not improved	Fast progress	No reduction	Decentralized	Implemented	Implemented	Implemented
P13	Green space development	Cropping pattern with low and medium water demand	Vegetation cover with low and medium water demand	Improved	Fast progress	Consumption control	Decentralized	Implemented	Implemented	Implemented

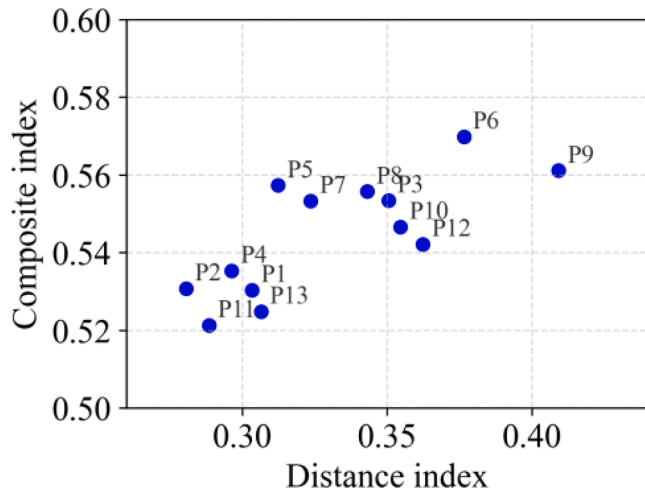


Fig. 4. Average performance of management packages during the simulation period.

3.4. Trade-offs and synergies

The concept of correlation was used to identify the relationships among the indicators. Fig. 7 shows the changes in pairwise correlation coefficients for the evaluated indicators under the superior management packages and the BAU scenario. Fig. 7a and b show that the relationships between the indicators in the two superior management packages are identical, although a slight difference in the correlation intensity between the indicators can be noted. Specifically, there is a trade-off between the NPC (economic cost of operating the system and implementing infrastructure projects) and five other indicators, namely R, GEP, UI, UME, WRE, and the distance index, under both superior packages. However, the relationships between the indicators vary in type and intensity under the BAU scenario. Fig. 7c indicates that the changes in the distance index under the BAU scenario are not significantly correlated with the two superior packages, and their values change relatively independently. Under the BAU scenario, the NPC index maintains a trade-off relationship with the R, GEP, UI, and UME indices. However, because there is insufficient investment in wastewater treatment and bioenergy production within this scenario, the WRE shows a synergy with economic costs. This means that as the economic costs of the system have increased annually, the inefficiency of the wastewater reclamation system has also increased. Additionally, under

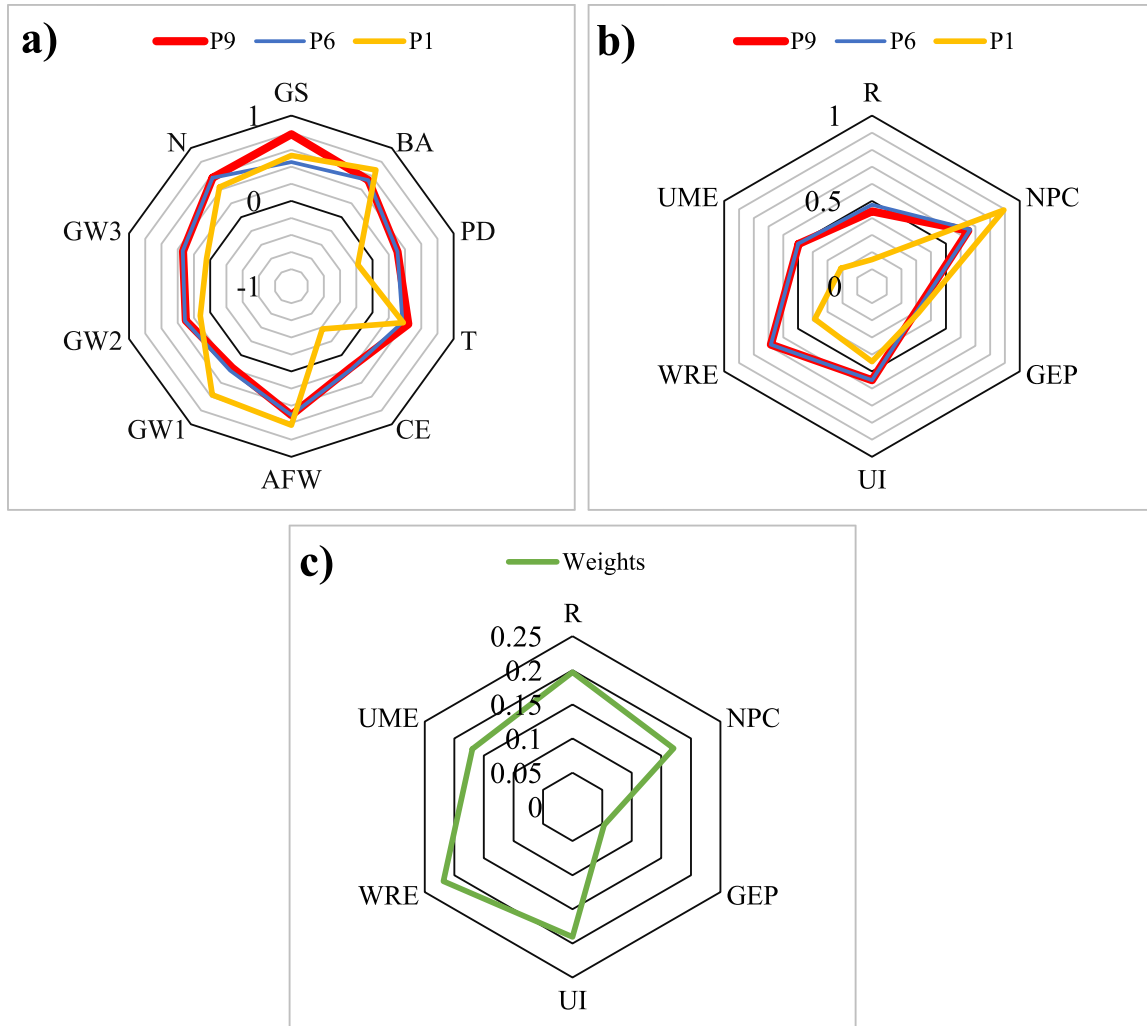


Fig. 5. The calculated average values for a) the indicators of the distance index and b) the sub-indices of the composite index assuming the implementation of the superior packages (P9 and P3) and the BAU scenario (P1) and c) the estimated weights for the sub-indices of the composite index (R: Reliability, NPC: Nexus policy cost, GEP: Gross ecosystem product, UI: Urban intensity, WRE: Water reclamation efficiency, UME: Urban metabolism efficiency) (Abbreviations in Figure (a) are defined in Table A2 in the appendix).

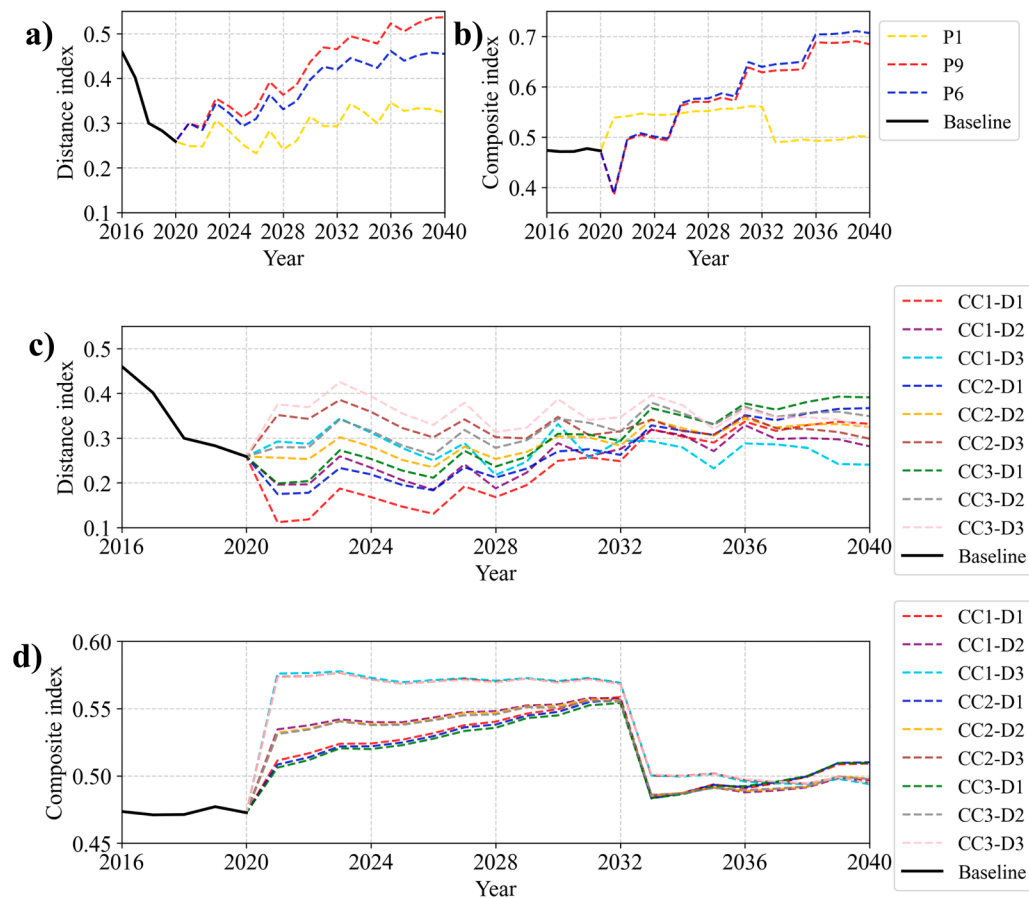


Fig. 6. The calculated average values in the base period (2016–2020) and the policymaking period (2021–2040) for a) the distance and b) the composite indices assuming the implementation of the packages P9, P6, and P1, and changes in the values of P1 in these two periods for c) the distance and d) the composite indices under demographic-climatic scenarios.

the BAU scenario, the WRE index has a trade-off relationship with other indicators and has not experienced growth, unlike the other indices.

3.5. Changes in the WFEES nexus components

The implementation of each management package has a specific impact on the components of the WFEES nexus. In this section, the results of the impacts are presented separately for each component, assuming the implementation of the superior management packages discussed in the previous section or the occurrence of the BAU management scenario.

3.5.1. Ecosystem services

This study assessed seven ecosystem services: water yield, stormwater retention, crop production, bioenergy production potential, carbon sequestration, urban cooling, and recreational spaces. Figure S2 in the online supplementary material shows the status of these services during the simulation period under different management packages. Two services in which precipitation is the determining variable—water yield and stormwater retention—showed similar trends during the period and under different packages. The average volumetric amount of water yield under package P1 was estimated to be 9.17 % and 3.37 % higher than that under packages P9 and P6, respectively, whereas the average volumetric amount of stormwater retention for package P9 was calculated to be 2.16 % and 0.42 % higher than that for packages P6 and P1, respectively. Since bioenergy production from agricultural residues relies on crop production, the trends of these two services are similar. The average crop production values were almost the same under packages P9 and P6 and were calculated to be 5.28 % lower than those under

P1. Additionally, the average potential bioenergy production under P1 was estimated to be 7.83 % higher than that under the other two packages. Another service is urban cooling, which is measured by the average perceived air temperature during the three warmest months of the year. The average temperature is estimated to be 0.3 % lower under P9 than under P1 and 0.34 % lower compared to P6. Carbon sequestration and recreational spaces have shown linear behavior because of the linear modeling of annual land use changes in this study. The estimated carbon sequestration rates are 3.65 % and 2.06 % higher under the current trend than those under packages P9 and P6, respectively. Conversely, the per capita green space in the BAU scenario is 18.7 % lower than that in P9 but 7.65 % higher than that in P6.

3.5.2. Water supply and demand

One part of the integrated simulator estimates the water demands and the amount of water allocations. Tables S9, S10, and S11 in the online supplementary material present the annual water supply and demand for each sector. According to the results, citizens and farmers have the highest water demand and, consequently, the highest water allocation. During the simulation period, the average water demand in this region under the BAU scenario (P1) will be 0.24 % and 1.31 % higher, respectively, compared to when P9 and P6 are implemented. Also, the average water allocation for all sectors under P1 is estimated to be 0.43 % lower and 0.22 % higher than those under P9 and P6, respectively.

In addition to assessing supply and demand, the quantitative and qualitative state of groundwater resources in this region, as the most important water supply resource, has also been assessed. Figs. A2 and A3 in the appendix provide information about the changes in water table

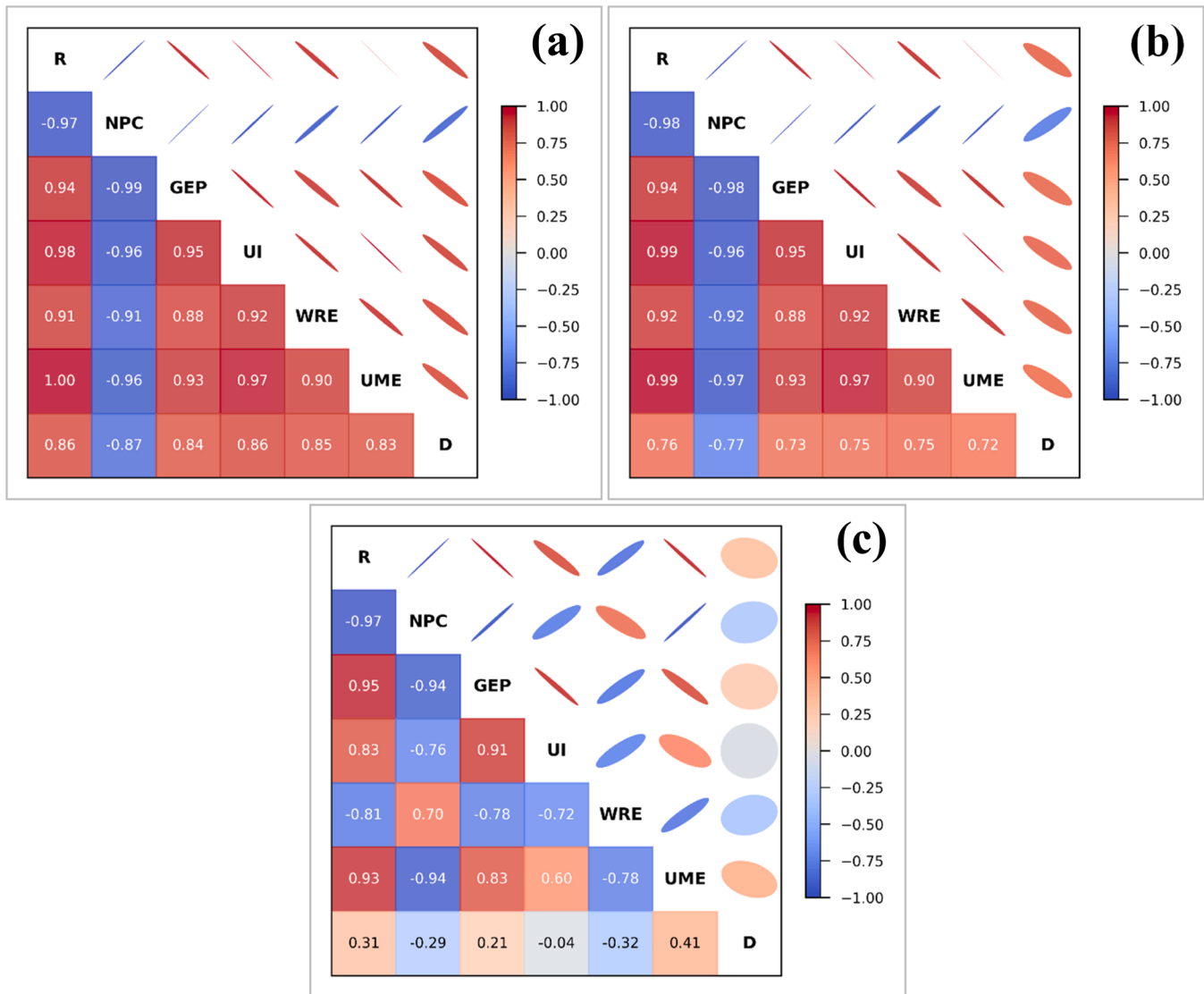


Fig. 7. The type and intensity of correlation for each pair of indicators under the implementation of management packages, a) P9, b) P6, and c) P1 (BAU scenario).

levels and nitrate concentrations in the study area's three aquifer zones. With the implementation of packages P9 and P6, the average water table levels in the second and third zones of the aquifer are expected to be higher than those in the BAU scenario. Additionally, the average nitrate concentration in zones 1 and 2 will be lower when these superior management packages are adopted than if the current trend continues. On the other hand, under package P1, the groundwater level in the first zone of the aquifer and the nitrate concentration in the third zone will be in better condition than if the superior packages were applied.

3.5.3. Urban water metabolism

The urban water metabolism section of the newly developed simulator focuses on energy, greenhouse gas, and material fluxes. Fig. 8 provides a summary of the outputs of this section of the simulator under the management packages. In general, superior management packages can reduce the negative impacts of urban metabolism and improve its efficiency compared to the BAU scenario. For example, the average energy consumption under the BAU scenario is estimated to exceed the consumption expected with the implementation of packages P9 and P6 by 0.62×10^7 kWh and 1.6×10^7 kWh, respectively. Also, the average annual greenhouse gas emissions under package P1 will be 5.74×10^7 and 5.93×10^7 kg-CO₂-equivalent higher than those under packages P9 and P6, respectively. Notably, the wastewater treatment rates of

packages P9 and P6 are higher than those of package P1. Consequently, the amount of agricultural fertilizer production from sludge and bio-energy production from biogas produced in wastewater treatment plants is estimated to be higher under these two packages (Fig. 8c and d).

If the superior packages are implemented, agricultural machinery and the maintenance of farms and green spaces constitute the largest share of future energy consumption. Following this category, energy consumption for groundwater pumping and the embodied energy in materials will rank second and third, respectively. If the current trend continues, although the energy used for machinery and maintenance is expected to be lower than that of superior packages, it will still represent the largest share of total energy consumed at 33 %. Additionally, the energy consumed from materials will be higher, making up 31 % of total energy consumption, whereas the energy used in pumping groundwater will account for 30 %. In terms of annual greenhouse gas emissions, sewage wells account for the largest share of annual greenhouse gas emissions across all the cases examined and under all three packages. If current trends continue, the average emissions from these wells will be 7.2×10^7 kg-CO₂-equivalent higher because of the lower rate of wastewater treatment compared with packages P9 and P6. The emissions from energy used for pumping groundwater and the embodied emissions in the materials will rank next in total annual emissions.

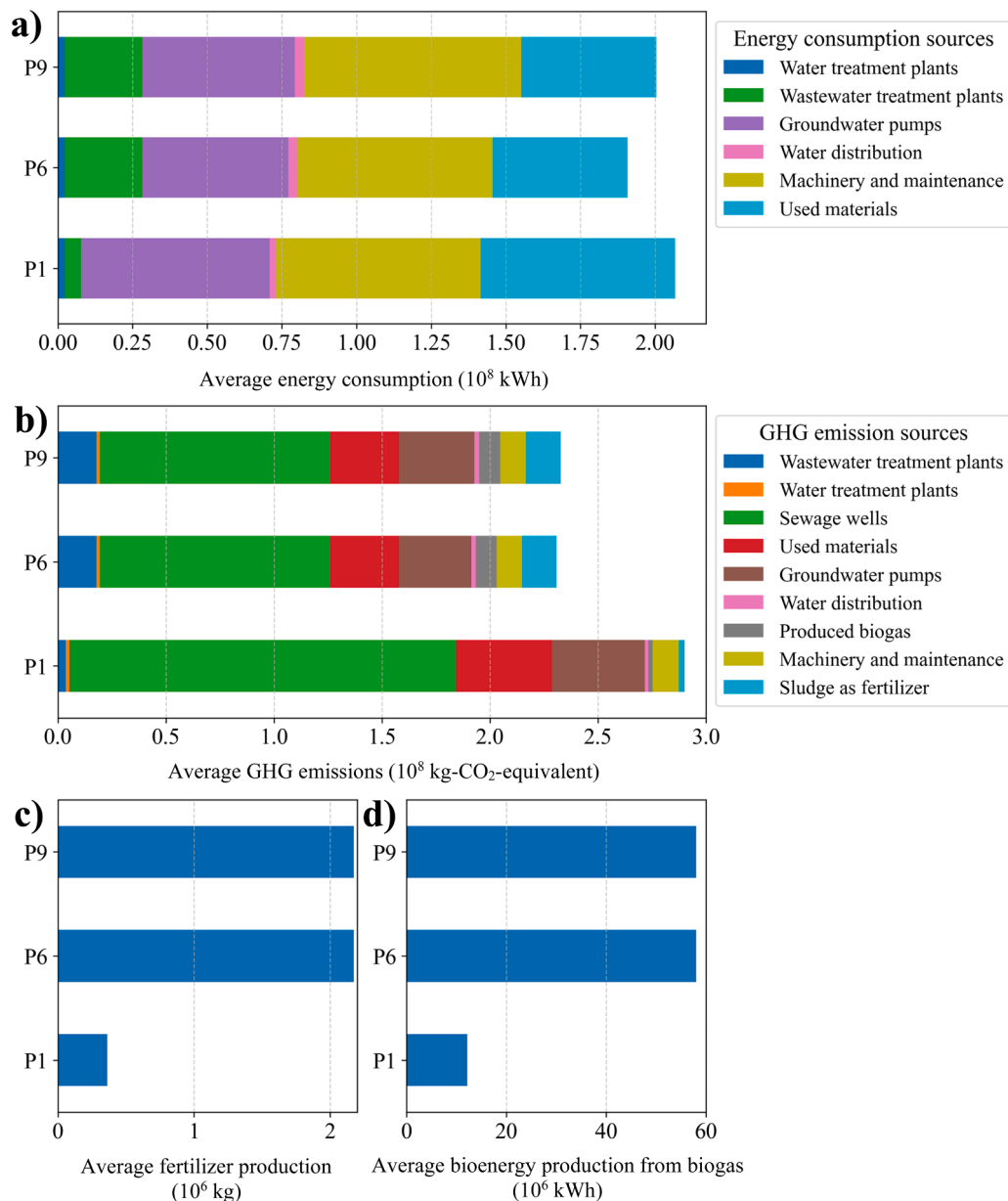


Fig. 8. A summary of the results from the urban water metabolism section (i.e., a) average energy consumption, b) average greenhouse gas emissions, c) average fertilizer produced from sludge, and d) average bioenergy production from biogas) under the superior management packages (P9 and P6) and the BAU scenario (P1).

4. Discussion

4.1. Interpretation of results

The integrated model developed in this study was run for a 25-year simulation period (2016–2040) and considered possible futures. The system's state was assessed based on defined indices (distance index and composite index). The possible futures included nine demographic-climatic scenarios and 13 management packages developed from combinations of different strategies. Concerning the demographic-climatic scenarios, an analysis of the distance index and composite index revealed that the CC3-D3 scenario—characterized by a combination of pessimistic population growth and the SSP5–8.5 climate change scenario—had the most detrimental effects on the system in both the short- and mid-term. Additionally, in the later years, the CC3-D1 scenario is projected to lead to slightly worse outcomes than other potential futures for the system (Fig. 6). This study aims to assess various management strategies to ensure that, regardless of which demographic-climatic

scenario unfolds, the region will adopt the most effective management approach in the future.

The management strategies considered were generally categorized into three types: 1) water use management, 2) implementation of infrastructure projects, and 3) a combination of water use management with infrastructure project implementation. Combining these three types of strategies with four different land use change scenarios, plus one package defined for the BAU scenario, resulted in 13 management packages. Fig. 4 illustrates the performance of these packages compared to one another, with packages P9 and P6 outperforming the other options. Both of these packages are based on the second type of strategy, which involves implementing infrastructure projects. The distinction between them lies in their approach to land use management. The findings indicate that projects such as constructing decentralized wastewater treatment plants and implementing artificial aquifer recharge will be more effective than water use management solutions, such as improving irrigation efficiency and utilizing vegetation with low to medium water requirements, though they may come at a higher cost.

The performance of packages P10 and P7, which focus on simultaneously implementing both infrastructure projects and water consumption management, closely matches that of the packages centered on water consumption management (P8 and P5), making them less desirable options.

Compared with the BAU scenario, the superior packages are projected to result in a lower average water demand in the future. Additionally, ecosystem services such as stormwater retention, recreational spaces, and urban cooling are expected to be in better condition when these packages are implemented than under the BAU scenario. On the other hand, other ecosystem services, such as water yield, carbon sequestration, crop production, and bioenergy production from agricultural residues, will be in a better state if the current trend continues. Groundwater resources are estimated to be in a better quantitative and qualitative state with the implementation of superior packages than with the BAU scenario. Sadri-Shojaei et al. (2025) investigated the effects of various management measures on ecosystem services and water resources in the future. They concluded that effective land use management focused on short-term urban land development, combined with the balanced long-term development of urban areas and green spaces, would improve groundwater resources and provide better ecosystem services in the future compared to current trends. Implementing projects such as wastewater treatment and artificial aquifer recharge would further increase these benefits. They suggested implementing water use management measures alongside infrastructure projects. However, our study findings indicated that focusing solely on infrastructure projects would be a more effective approach than combining them with consumption management solutions. In another study, which focused on the same region as this paper, Emamjomehzadeh et al. (2023) examined future urban water metabolism under these measures and solutions. Similar to our findings, they demonstrated that constructing decentralized wastewater treatment plants could lead to lower energy consumption and greenhouse gas emissions than those achieved in the BAU scenario.

As shown in Fig. 3, the connections between urban water metabolism, ecosystem services, and water resources can be understood through the WFEES framework. Consistent with previous studies, the findings indicate that there are significant dynamics between urban metabolic flows and ecosystem services under various social-environmental scenarios (Cárdenas-Mamani & Perrotti, 2024). Ecosystem services, such as carbon sequestration and bioenergy production, can effectively reduce carbon emissions generated by metabolic processes and help offset the energy consumption associated with these processes (Yusuf et al., 2025). Additionally, another crucial ecosystem service, which involves regulating both water quantity and quality, not only provides water to consumers but also reduces energy and material consumption, as well as greenhouse gas emissions related to water supply and treatment within the urban water metabolism sector (Cárdenas-Mamani & Perrotti, 2022). Moreover, how urban metabolic flows are managed has a direct impact on ecosystem services. For instance, producing fertilizer from waste generated at urban wastewater treatment facilities contributes positively to the ecosystem service of crop production. The results demonstrate that enhancing the efficiency of urban metabolism, along with improved management of resources and consumption, leads to better outcomes for ecosystem services.

4.2. Prospect applications

Effective water governance is a socio-political and environmental challenge that extends beyond economic considerations, requiring a multi-dimensional policy approach. The absence of integrated governance strategies can fuel a vicious cycle of overextraction, destructive competition, and the degradation of common-pool resources, ultimately jeopardizing long-term sustainability. To break this cycle, policymakers must prioritize institutional strengthening, participatory governance, and adaptive resource management policies that are both consistent

with regional ecological thresholds and socially equitable (Momeni et al., 2023). The findings of this paper can be practically applied to long-term planning within the study area. These findings provide recommendations for regional policymakers regarding land management and the implementation of projects such as wastewater treatment, bio-energy production from biogas produced in wastewater treatment plants, artificial aquifer recharge, and rainwater harvesting. Additionally, the methodology and simulation tool developed in this study will benefit various organizations, including municipalities, water resource management agencies, environmental protection organizations, and researchers looking to undertake similar studies in other regions. The methodology developed in this study focuses on the interactions between facilities and human activities in urban water metabolism and the provision of ecosystem services. This approach significantly contributes to reducing emissions, as well as energy and material consumption. Additionally, it supports the optimal management of land and water resources in urban areas. The integrated simulator serves as a robust instrument for urban planning. It considers various factors, including climate change, land use changes, and population growth. This tool utilizes economic, environmental, and social indicators while emphasizing the importance of resilience thresholds.

4.3. Limitations

Notably, there were limitations in the development of the simulation model for this study. This model performs calculations annually and cannot simulate monthly or weekly variations. The land use and population changes analyzed in this study are assumed to be based on historical trends and are considered linear. Most parameters required for the ecosystem services and urban water metabolism models were selected using values similar to those suggested by previous studies in comparable areas rather than being directly measured. Additionally, the indicators introduced are based on the available data and the outputs from the newly developed simulation model. In the economic section, the inflation rate is based on average global inflation and is assumed to remain constant. This assumption may lead to overestimating or underestimating costs and benefits in the calculations. Although the simulation outputs related to surface water, groundwater, and key ecosystem services (e.g., water yield, stormwater retention) were validated using available observed data, other components of the urban metabolism module—such as energy consumption, greenhouse gas (GHG) emissions, and material flows—could not be empirically validated due to the unavailability of corresponding observed datasets. This limitation stems from the lack of consistent and disaggregated monitoring data at the appropriate spatial and temporal scales. This limitation is acknowledged to ensure transparency and should be addressed in future research as more empirical data becomes available to support comprehensive model validation. While temporal analysis was conducted in this study, spatial analysis was not performed across different regions because of the small study area, which may lead to generalized estimates. To enhance the precision and applicability of future research, a higher-resolution spatial approach should be employed. This would allow for a more detailed assessment of regional variations in resources and environmental governance challenges, resource distribution, and ecosystem responses, ultimately leading to more targeted and effective policy recommendations.

5. Concluding remarks

The concept of the WFEES nexus is highly relevant to the development of comprehensive urban planning and policymaking related to natural resources and infrastructure. This paper presented a comprehensive tool for simulating the various components of the nexus by developing a novel integrated simulator consisting of ecosystem services, urban water metabolism, end users, and regional water resources. This tool modeled the annual interactions between the components and

generated the necessary outputs for urban water and environmental management. The study also proposed a novel composite index comprising various sub-indices, including reliability, gross ecosystem production, management policy costs, urbanization intensity, water reclamation efficiency, and urban metabolism efficiency. In addition to these indices, the regional safe operating space framework, which measures the state of the system compared with standards and thresholds, was also utilized.

To simulate the future of this nexus, nine demographic-climatic scenarios and 13 management packages featuring different strategies were developed. The integrated simulator was run for 25 years (2016–2040) and simulated the state of the defined system under 117 possible futures. The findings showed that, compared with the BAU scenario, the best management packages under all demographic-climatic scenarios would perform better on average in terms of the quantitative and qualitative state of groundwater, several ecosystem services, and the efficiency of urban water metabolism. These packages estimated higher values for the composite and the distance-to-threshold indices on average compared to the BAU scenario. According to these findings, policymakers in the region should prioritize the implementation of infrastructure projects such as completing the wastewater collection network, building decentralized sewage treatment plants, and providing artificial aquifer recharge. These projects effectively reduce the energy consumption for pumping groundwater and the GHG emissions from this energy use. They help decrease emissions from wastewater wells, control groundwater depletion, and produce fertilizer for the agricultural sector. Additionally, the bioenergy production project, which utilizes biogas generated from wastewater treatment plants, not only helps reduce GHG emissions but also provides a portion of the energy needed for the system. The rainwater harvesting project also decreases the demand for non-potable water in urban areas. Regarding land management, the top packages proposed two approaches: urban land development and the balanced development of urban land and green spaces. Although these solutions are more costly than the BAU scenario and water consumption management policies, they are strongly recommended based on indicators such as system efficiency, reliability,

distance from resilience thresholds, and urban well-being.

It is recommended that similar future studies that combine the WFEES linkage concept with urban planning take into account stakeholders' interests and opinions when selecting superior management strategies. Identifying key stakeholders, including policymakers, local communities, industries, and environmental organizations, will help develop more inclusive and effective governance frameworks. Also, for urban areas facing high population growth and urbanization, other ecosystem services such as urban flood risk mitigation, noise pollution reduction, air pollutant removal, and access to nature can be considered in simulations and decision-making.

CRediT authorship contribution statement

Sina Sadri-Shojaei: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Marzieh Momeni:** Writing – review & editing, Writing – original draft, Visualization, Software, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Reza Kerachian:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Conceptualization. **Zahra Kalantari:** Writing – review & editing, Writing – original draft, Validation, Methodology. **Omid Emamjomehzhadeh:** Visualization, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.scs.2025.106781](https://doi.org/10.1016/j.scs.2025.106781).

Appendix

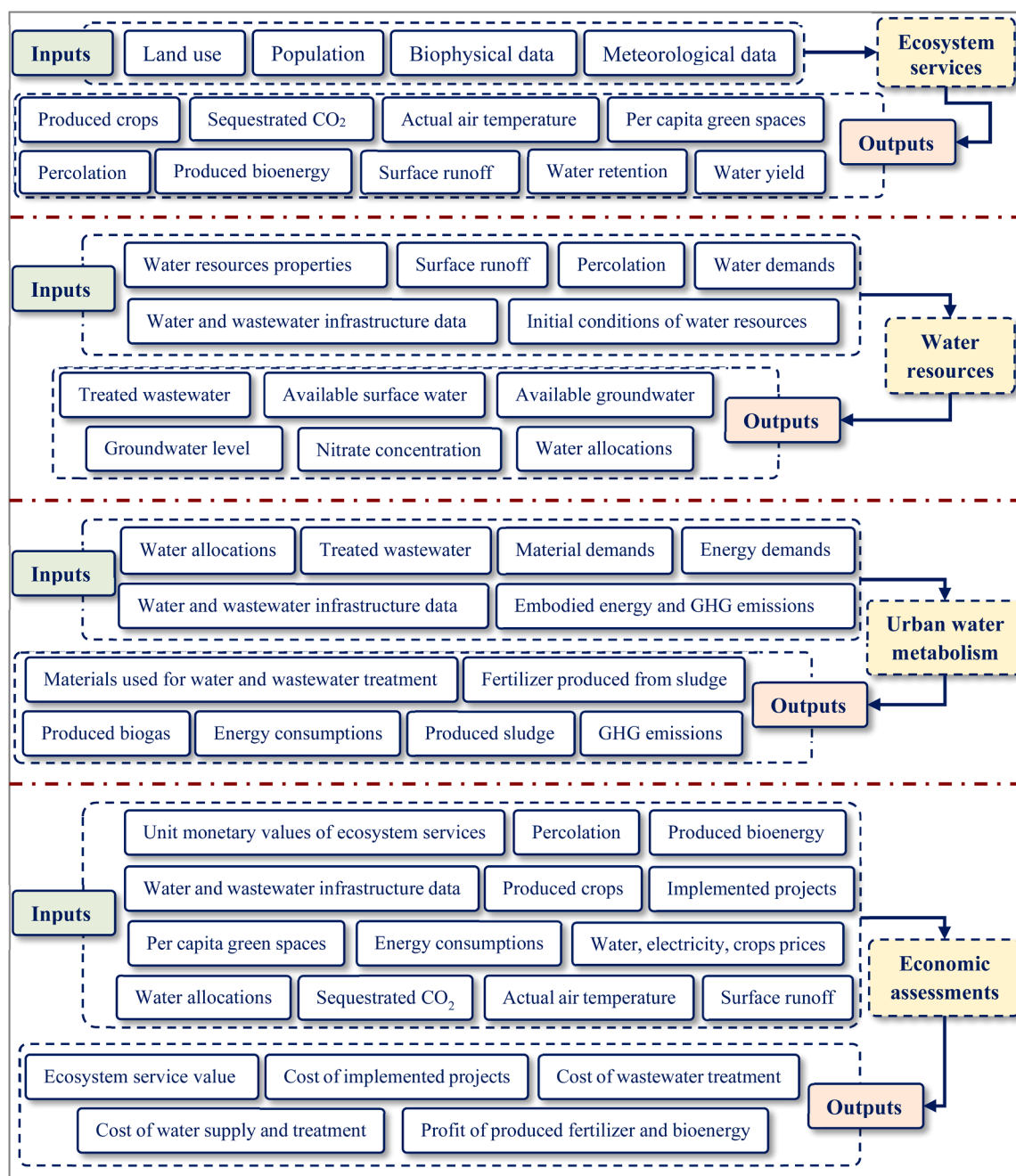


Fig. A1. A summary of integrated simulator models and their outputs and inputs.

Table A1

Groundwater pumping pricing details for different consumers (Skardi et al., 2020).

Sector	The cost of pumping a cubic meter of water per meter for each consumer in the base year (2016) (USD/m ³ /m)
Agriculture	0.003
Industry	0.018
Urban	0.007

Table A2
Defined indicators and thresholds within the RSOS framework.

Sector	Indicator (code)	Limit	Reference
Land use system	Per capita green space (GS)	$\geq 7 \text{ m}^2/\text{capita}$	Bahrini et al. (2017)
	Built-up lands (BA)	$\leq 70\%$	Tehrani and Makhdoum (2013)
	Population density (PD)	$\leq 110 \text{ person/ha}$	Tehrani and Makhdoum (2013)
Climate	Average maximum air temperatures during the warm months (T)	$\leq 45^\circ\text{C}$	Vecellio et al. (2022)
	CO ₂ emissions per capita (CE)	$\leq 2.3 \text{ ton/person}$	Gore (2021)
Freshwater availability	Freshwater availability per person for urban demand (AFW)	$\geq 36.5 \text{ m}^3/\text{person/year}$	WHO (2003)
Groundwater	Water table level in aquifer (zone 1, zone 2, zone 3) (GW1, GW2, GW3)	$\geq 1117.8 \text{ m}, \geq 1038.2 \text{ m},$ $\geq 989.1 \text{ m}$	Tehran Regional Water Company
		$\leq 50 \text{ mgNO}_3/\text{L}$	Panahi and Alavi Moghaddam (2012)
Nutrient cycle: Nitrogen	Nitrate concentration in groundwater (N)		

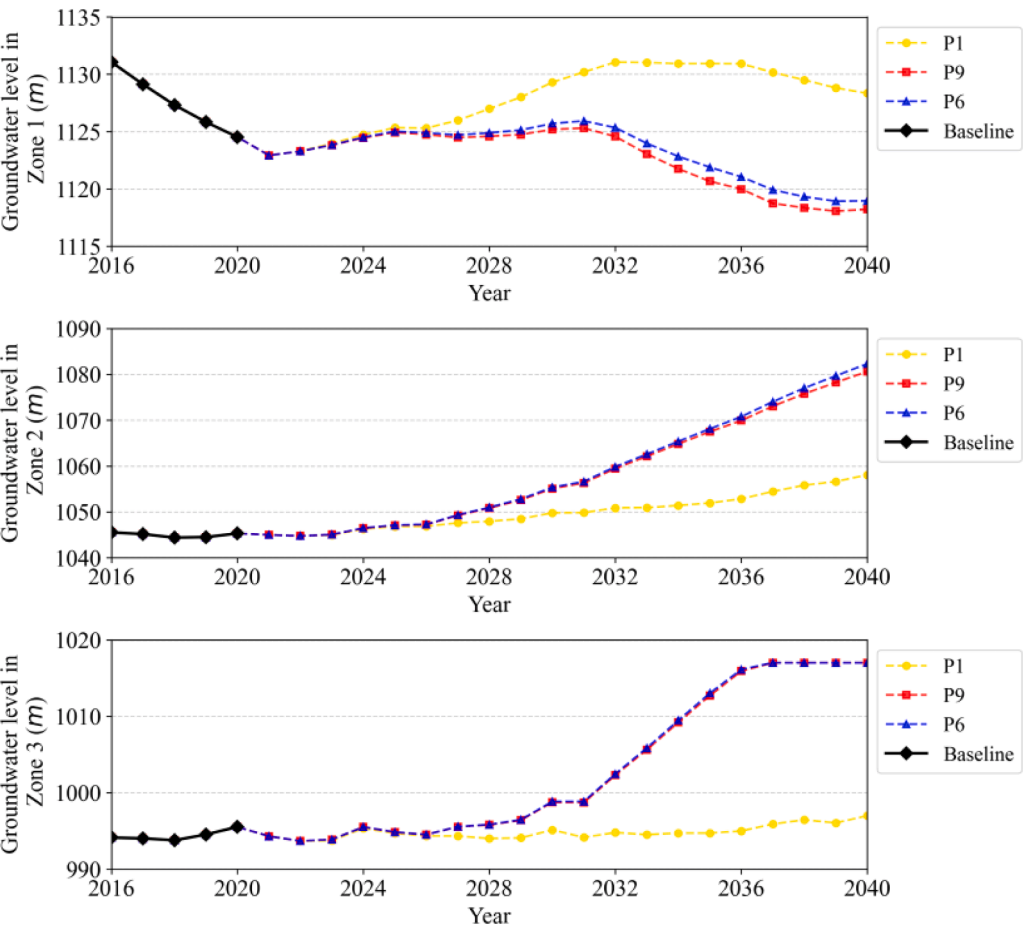


Fig. A2. A comparison between the impacts of the superior management packages (P9 and P6) and the BAU scenario (P1) on groundwater levels in different zones of the regional aquifer.

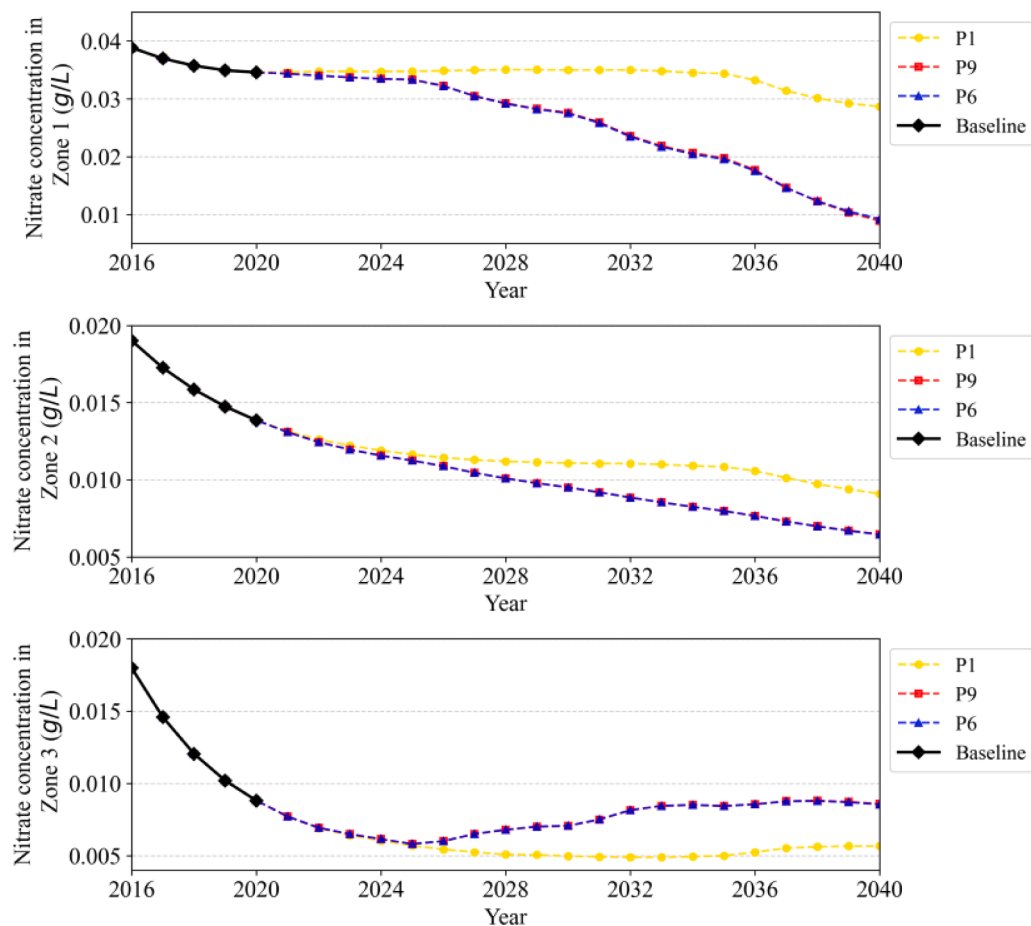


Fig. A3. A comparison between the impacts of the superior management packages (P9 and P6) and the BAU scenario (P1) on nitrate concentration in different zones of the regional aquifer.

Data availability

Data will be made available on request.

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